Rishant Prakash

rpraka15@asu.edu MS in Geological Sciences,
Arizona State University

Research Interests: Cosmochemistry, Origin and Volatile Depletion of the Solar System, High & Low Temperature Geochemistry, Mass Spectrometers

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Education

M.S. Geological Sciences	GPA: **3.9/4.0**
– Arizona State University	(Aug 2023-present)
MTech. Earth Sciences	GPA: **7.9/10**
– Indian Institute of Science, Bengaluru	(2021-23)
M.Sc. Geology	GPA: **8.32/10**
– Hansraj College, University of Delhi	(2019-21)
B.Sc. Geology	GPA: **7.46/10**
– Hansraj College, University of Delhi	(2016-19)

Research Projects Completed/Ongoing

- **Origin and Distribution of Ti nucleosynthetic anomalies in the Solar Nebula** *Aug 2023 – present*
  Prof. Meenakshi Wadhwa School of Earth and Space Exploration (ASU)
  - Developing and refining in-situ analytical protocols for high-precision Ti isotope measurements of diverse CAIs (including FUN CAIs) using LA-MC-ICPMS.
  - Synthesizing and documenting Ti-doped CMAS glasses with varied V and Cr ratios to serve as internal standards for Ti isotope analyses, employing a Gas-Mixing furnace at the University of Arizona, Tucson
  - Worked on clean laboratory procedures for Mg purification in achondrites, including column chemistry, acid titration, and hazardous waste management in Class 1000 cleanroom facilities.

- **Modeling nucleosynthetic signatures of presolar grains using 3D core-collapse supernova explosion (CCSNe)** *Aug 2024 – present*
  Prof. Patrick Young, Prof Larry Nittler School of Earth and Space Exploration (ASU)
  - Working with 3D CCSNe model to reproduce isotopic signatures of Ti & Cr observed in presolar oxide grains.
  - Running numerical simulations to examine the role of explosive nucleosynthesis, mixing, and fallback in producing observed isotopic anomalies.
  - Comparing simulation outputs with laboratory measurements of presolar grains (nano-SIMS datasets) to establish astrophysical sources of isotopic reservoirs.

- **Acid Leaching Experiments of Silicate rocks and its implications for the mobility of cations during silicate weathering and dissolution of extra-terrestrial samples** *August 2021 –2023*
  Prof. Ramananda Chakrabarti Centre for Earth Sciences (IISc, Bangalore)
  - Conducted acid leaching experiments on various silicate rock types (ultramafic to felsic) to understand cation mobility during silicate weathering, utilizing non-HF acids and High-Pressure Asher (HPA) digestion for expedited reactions.
  - Investigated the correlation between silica content and residue weight in silicate rocks, demonstrating mineral stability order from felsic to ultramafic without the use of HF; REE and major element behavior in leachate and residue
  - Performed geochemical characterization of the Sadiya meteorite to classify it as an H/L chondrite based on moderately siderophile elements (Ni, Co, Fe) from bulk composition data. ***Thesis Link***.

- **Geochemical analysis of Ameboidal Olivine Aggregates (AOAs) in CM and CV chondrites**
  Summer Intern Project Advisor: Dr. Dipak Panda *May 2022 – July 2022*
  Planetary Science Division, PRL, Ahmedabad
  - Conducted comprehensive geochemical analysis of Ameboidal Olivine Aggregates (AOAs) in Mukundpura CM and Allende CV chondrites using EPMA (EDS and WDS) techniques in JEOL 8530F FE-EPMA.
  - Documented significant aqueous alteration in Mukundpura CM2.5 through abundant phyllosilicates, carbonates and PCPs, contrasting with the relatively unaltered Allende CV3.

- Interpreted the presence of low-Ca pyroxene on AOA peripheries, indicating condensation in a ^{16}O-poor gaseous nebula, linking AOA evolution to chondrule formation.
- Abstract presented online at MetMESS 2022. **Abstract Link, Report Link** .

· **Land Use and Land Cover map of Dehradun City and accuracy assessment over the year 2001-2021 using Remote Sensing and GIS** 2020-2021

M.Sc. Dissertation Prof. Vimal Singh *Department of Geology, University of Delhi*

- Utilized Landsat satellite imagery (Landsat 7 ETM, Landsat 4-5 TM, Landsat 8 OLI/TIS) for LULC classification across 2001,2011 and 2021; Executed supervised classification using the Maximum Likelihood Classifier in ERDAS IMAGINE 2014
- Conducted rigorous accuracy assessments for LULC classifications supported by error matrices and Kappa statistics; Quantified and interpreted significant spatio-temporal land use transformations. **Dissertation Link**

· **Interpretation of Depositional Environment from Lithofacies and Well Log Analysis in KG Basin**

2019

Industry Internship Advisor: GM. Rajesh Sharma Oil and Natural Gas corporation (ONGC), Dehradun

- Interpreted depositional environments (fluvial to marginal marine) and lithofacies of synrift sequences in the Krishna-Godavari (KG) Basin using integrated well log analysis (gamma ray, resistivity), core data, and biostratigraphic reports from four wells.
- Characterized synrift sandstones by analyzing textural and mineralogical immaturity, diagenetic alterations and their implications for reservoir quality and fluid mobility; Correlated gamma ray log characteristics with specific sedimentological features to reconstruct paleoenvironments.
- Utilized biostratigraphic and palynological data to assign Late Jurassic to Early Cretaceous ages to the interpreted synrift sequences, establishing a robust chronological framework for basin evolution. **Report Link**

Geological Field Training

· **Holey Tour Arizona, USA** **Sept' *2023***

- Meteor Crater formation (overturned strata) and adjoining areas.
- Sunset Crater and adjoining areas- AA Lava Flow type ~1000 yr old eruption.
- Understanding of meteorite impact crater vs fluid interactions leading to crater.

· **Field Trip to Ajeetgarh, Rajasthan** **Dec' *2019***

- Mapped the Precambrian geology of Ajeetgrah and adjoining areas.
- Prepared the systematic geological maps of the greater Ajeetgarh area.
- Studied the structures of the area and cross-beddings in basal Alwar Group.
- Identified the major sequence of structural deformations (foliation and lineation mineralization)

· **Field Trip Dehradun-Mussoorie (Himalayan Frontal/Boundary Thrust), Uttarakhand, India**

Sept' 2018

- Study focused on structural and sedimentological features on a dry riverbed.
- Marking the major fault boundaries of the **Himalayan Frontal Thrust (HFT) & Main Boundary Thrust (MBT),** along with the drainage divide of river Ganga and Yamuna.
- Basic mapping techniques and use of geological instruments.

· **Field Trip to (Precambrian) Varanasi, Uttar Pradesh, India** **March 2018**

- Collected rock samples from Semri and Rewa formations of the Vindhyan Supergroup.
- Studied the sedimentary structures and attitude of a Proterozoic era non-perennial riverbed.
- Measured 1.5 Ga stromatolites in the Sulkan Valley.

Field Trip to Nainital, Uttarakhand, India **Dec' 2016**

- Studied the geological setting of Nainital Lake at an elevation of 1900 m above sea level in the Higher Himalayas (Above Shivaliks)
- Structural and sedimentological analysis of the Krol (neo-Proterozoic) and Tal Formation (early Cambrian).

Research Contributions:

Prakash, R., Young, P. (2026). Presolar Oxide Grains as probes for symmetric and asymmetric Core-Collapse Supernova Explosions. **247th American Astronomical Society** (AAS) 2025, #ID371	4-8 January, 2026
Campbell, A., **Prakash, R.**, Chakraborty, A., Vierling, S., Garvie, L., Wittmann, A. (2025). *Santa María del Oro (IIIAB, Iron), Durango, Mexico*. **Meteoritical Bulletin, No. 114.** (Link)	Approved July 2025.
Prakash, R., Rai, V.K., Aigner, M., Hines, R., Wadhwa, M. (2025) Origin and Distribution of Ti nucleosynthetic anomalies in the Solar Nebula. (SESE Symposium Poster)	22 Feb, 2025
Prakash, R. and Panda, D. (2022) Geochemical Analysis of Ameboidal Olivine Aggregates (AOAs) in Mukundpura (CM2) and Allende (CV3) chondrites. Presentation at Meteoroids, Meteors and Meteorites: Messengers from Space (MetMESS) 2022 (Abstract S1-09-Link)	24 - 25 November 2022

Positions of Responsibility

- **SESE Outreach Program:** Buseck Center for Meteorite Studies, ASU **2023-Present**
  Explained extraterrestrial and terrestrial rocks to people from various backgrounds.
- **IISc Open Day (Outreach):** Centre for Earth Sciences, ASU **2022-23**
  Interacted with people from various backgrounds and exposed them to meteorites and science experiments.
- **Coordinator,** American Association of Petroleum Geologists (AAPG) Student Chapter **2018–19**
  Organized and managed events, quiz competitions, and social media in Delhi University.
- **Technical Head,** PIXELS, The Photography Society of Hansraj College **2017-19**
  Led a team of 40 undergrads. Organized photo-walks, taught photojournalism and editing, and covered campus events.
- **Non-Profit Teacher,** NAV UMEED, NGO, North Delhi **2017-19**
  Helped high school students from marginalized backgrounds with math and science problems in the evening class.
- **Technical Head,** VISION, Hansraj College Newspaper Society **2017-18**
  Led editorial design and publication of a biweekly student newspaper for Hansraj College, Delhi University.
- **Non-Profit Volunteer,** National Service Scheme Rural Development Wing **2016-18**
  Facilitated workshops on personal hygiene, education value, and poverty alleviation for marginalized youth in North Delhi.

Awards & Scholastic Achievements

- **SESE Exploration Graduate Fellowship (SEG)-** School of Earth and Space Exploration, ASU **2026**
- **Graduate College Travel Award** -247th American Astronomical Society Meeting (AAS) **2026**
- **University Graduate Fellowship (UGF)-** The College of Liberal Arts and Sciences, ASU **2025**
- **Gate Scholarship** for pursuing two-year MTech at Indian Institute of Science, Bengaluru **2021-23**
- **Summer Research Fellowship** at Physical Research Laboratory, Ahmedabad **2022**
- Post Graduated **First Division with Distinction** in Masters in Geology, Hansraj College, Delhi University **2021**
- **International Geological Congress** (IGC) 2020 elected student volunteers under 300 all over India **2020**
- **2nd in National Geological Quiz** conducted by AAPG student Chapter Delhi University **2016-17**

Technical Skills

- **Analyses & Wet Lab**: LA/MC-ICP-MS (Neptune, Photon Machine 193 nm UV Laser), ICPMS (Thermo X-Series II) EPMA (JEOL), SEM (Phenom XL), Clean Lab 1000 & 10000.
- **Experimental**: Gas Mixing Furnace and High-Pressure Asher (HPA)
- **Software**: ArcGIS, Erdas Imagine, OriginPro, Google Earth, Adobe Suite, Microsoft Office, Google Suite
- **Programming**: Python, C++, LaTeX

Languages

- English, Hindi: Fluent